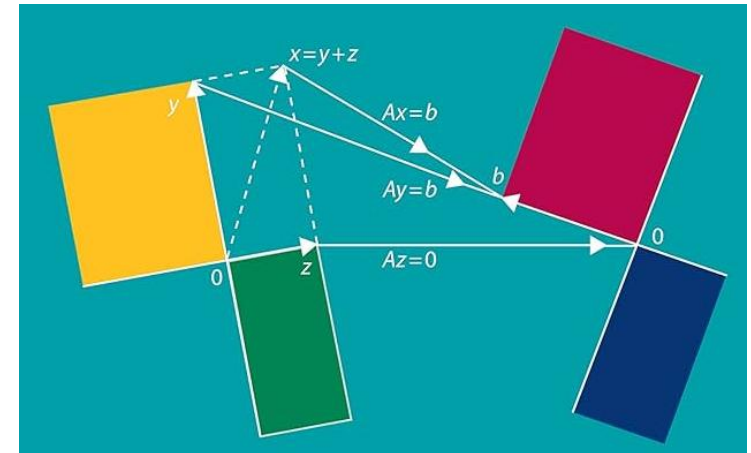


Principle Component Analysis (PCA)

(Linear Algebra)

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PCA

- Principal Component Analysis, is a statistical technique used to reduce the dimensionality of a dataset while preserving as much of the original variability as possible.
- PCA transforms the original data into a new set of variables called principal components.
- These principal components are linear combinations of the original variables and are ordered such that the first few retain most of the variation present in the original dataset.

PCA

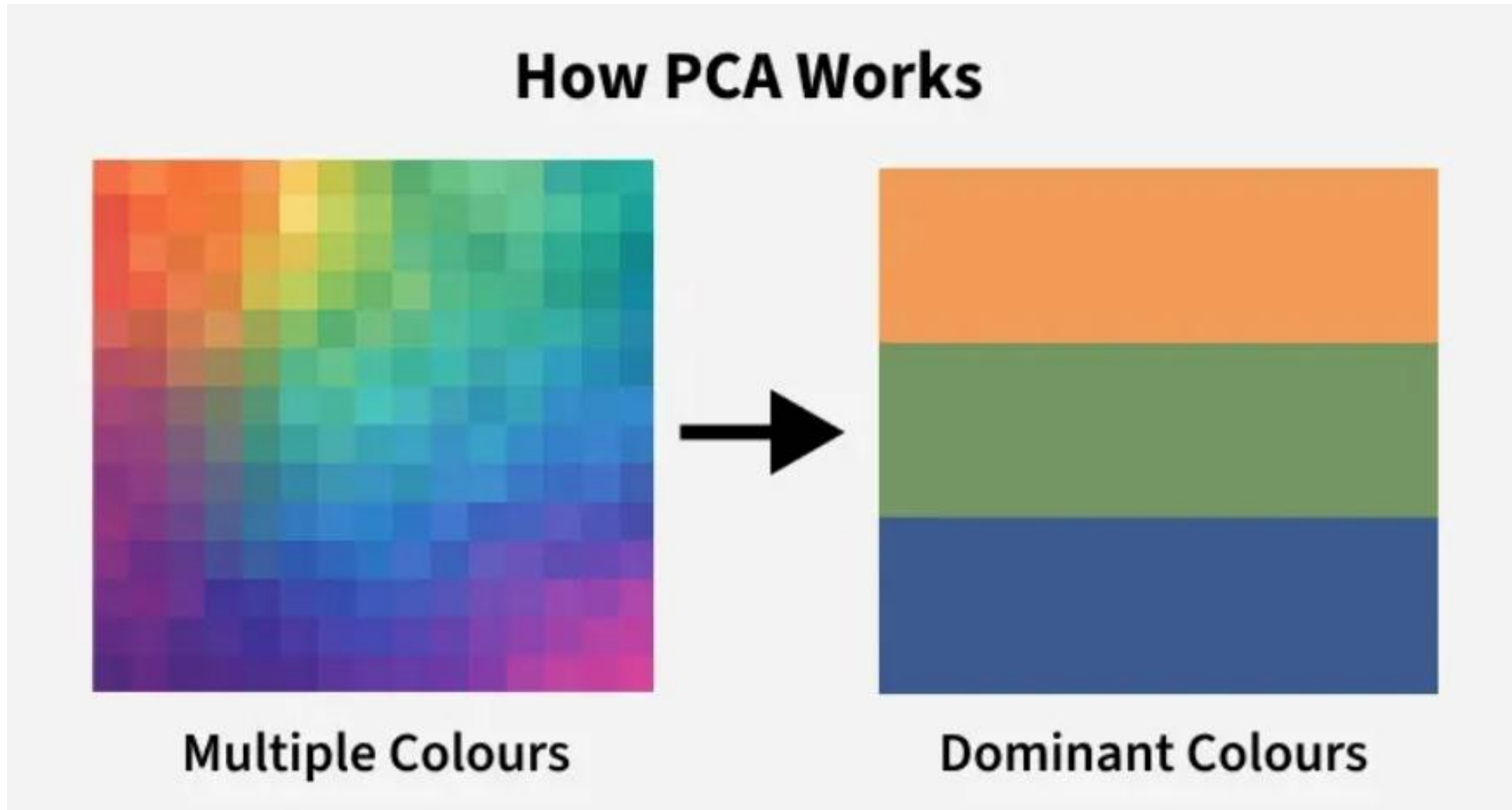
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From High-Dimensional Data to Meaningful Low-Dimensional Representations

Why PCA?

- It simplifies complex data by reducing the number of variables, making it easier to analyze and visualize.
 - **Dimensionality Reduction:** PCA reduces the number of variables in a dataset while retaining the most important information.
 - **Data Visualization:** By reducing data to 2 or 3 dimensions, PCA allows for easier visualization, helping to identify patterns, trends, or clusters in the data.
 - **Noise Reduction:** PCA can help remove noise by focusing on the components that capture the most variance, effectively filtering out less important information.
 - **Feature Extraction:** It helps in identifying the most important features in the data

Examples



PCA vs SVD

PCA: Involves finding the eigenvectors (principal components) and eigenvalues of the covariance matrix of the data. The eigenvectors correspond to the directions of maximum variance, and the eigenvalues indicate the amount of variance along each direction.

SVD: Decomposes any rectangular matrix A into three matrices: $A = U\Sigma V^T$, where:

- U contains the left singular vectors (orthonormal columns).
- Σ is a diagonal matrix with singular values (square roots of the eigenvalues).
- V^T contains the right singular vectors (orthonormal rows).

Principal Component Analysis

- **PCA**, or **Principal Component Analysis**, is an algorithm to reduce a large data set without loss of important information.
- **PCA** is defined as an orthogonal linear transformation that transforms the data to a new coordinate system such that the greatest variance by some scalar projection of the data comes to lie on the first coordinate (called the **first principal component**), the second greatest variance on the second coordinate, and so on.
- In simpler words, it detects the directions for maximum variance and project the original data set to a lower dimensional subspace (up to a change of basis) that still contains most of the important information.

Example

Consider a large dataset with m samples and 30 different cell features. There are many variables that are highly correlated with each other. We can create an $m \times 30$ matrix $\mathbf{A}$, with the columns $\mathbf{F}_i$ representing the different features.

$$A = \begin{bmatrix} \vdots & \vdots & \vdots \\ F_1 & \cdots & F_{30} \\ \vdots & \vdots & \vdots \end{bmatrix}$$

Now suppose we want to reduce the feature space. One method is to directly remove some feature variables. For example, we could ignore the last 20 feature columns to obtain a reduced data matrix $\mathbf{A}^*$. This approach is simple and maintains the interpretation of the feature variables, but we have lost the dropped column information.

$$A = \begin{bmatrix} \vdots & \vdots & \vdots \\ F_1 & \cdots & F_{30} \\ \vdots & \vdots & \vdots \end{bmatrix} \implies A^* = \begin{bmatrix} \vdots & \vdots & \vdots \\ F_1 & \cdots & F_{10} \\ \vdots & \vdots & \vdots \end{bmatrix}$$

Data Centering

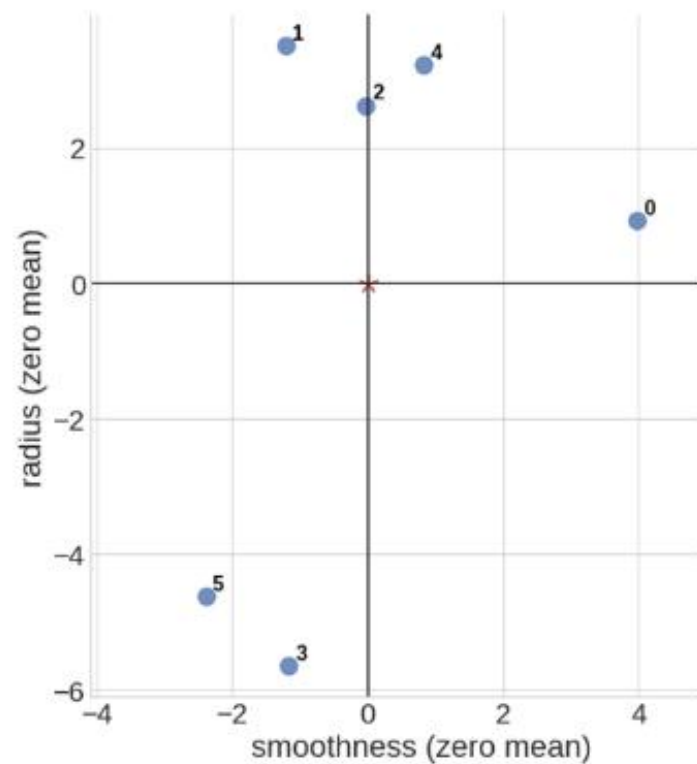
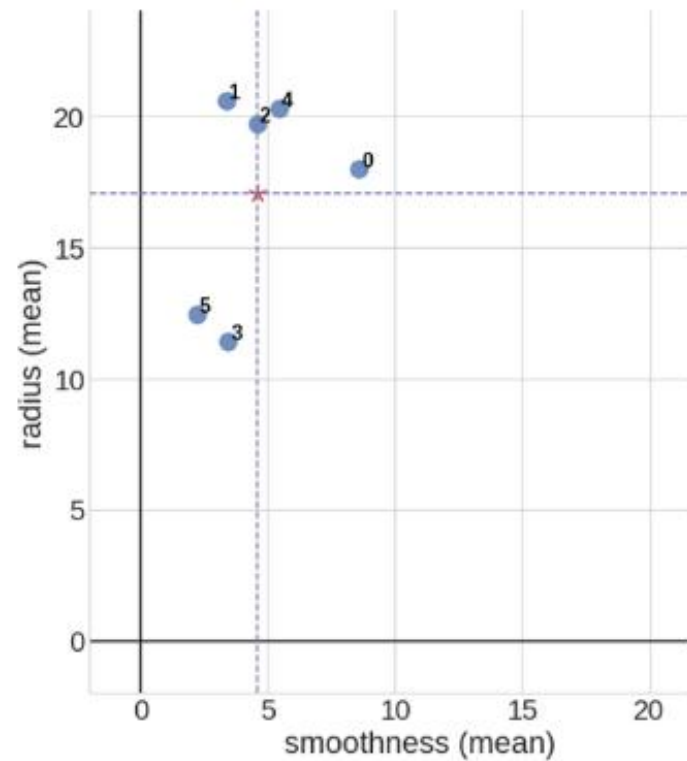
- The first step of PCA is to *center the data*.
- We carry out a data shift to our data columns of A such that each column has a mean of 0.
- For each feature column F_i of A , we calculate the mean $F_i\text{-Bar}$ and do a subtraction of $F_i\text{-Bar}$ from each entry of the column F_i . We do this procedure to each column, until we obtain a new centered data set A .

Example 🔗

This is an example of data centering for a set of 6 data points in 2d coordinate space:

$$p_0 = (8.6, 18.0), p_1 = (3.4, 20.6), p_2 = (4.6, 19.7), p_3 = (3.4, 11.4), p_4 = (5.4, 20.3), p_5 = (2.2, 12.4)$$

We first calculate the mean point $\bar{p} = (4.6, 17.1)$, then shift all of the data points such that our new mean point is centered at $\bar{p}' = (0, 0)$.



Covariance Matrix

For our *centered* data set $\mathbf{A}$ of dimension $m \times n$, where m is the total number of data points and n is the number of features, the *Covariance Matrix* is defined to be

$$Cov(\mathbf{A}) = \frac{1}{m-1} \mathbf{A}^T \mathbf{A}.$$

The diagonal entries of $Cov(\mathbf{A})$ explain how each feature correlates with itself, and the sum of the diagonal entries is called the overall variability (total variance) of the problem.

Covariance Matrix

Example

Consider a covariance matrix in the form below. From this matrix, we can obtain:

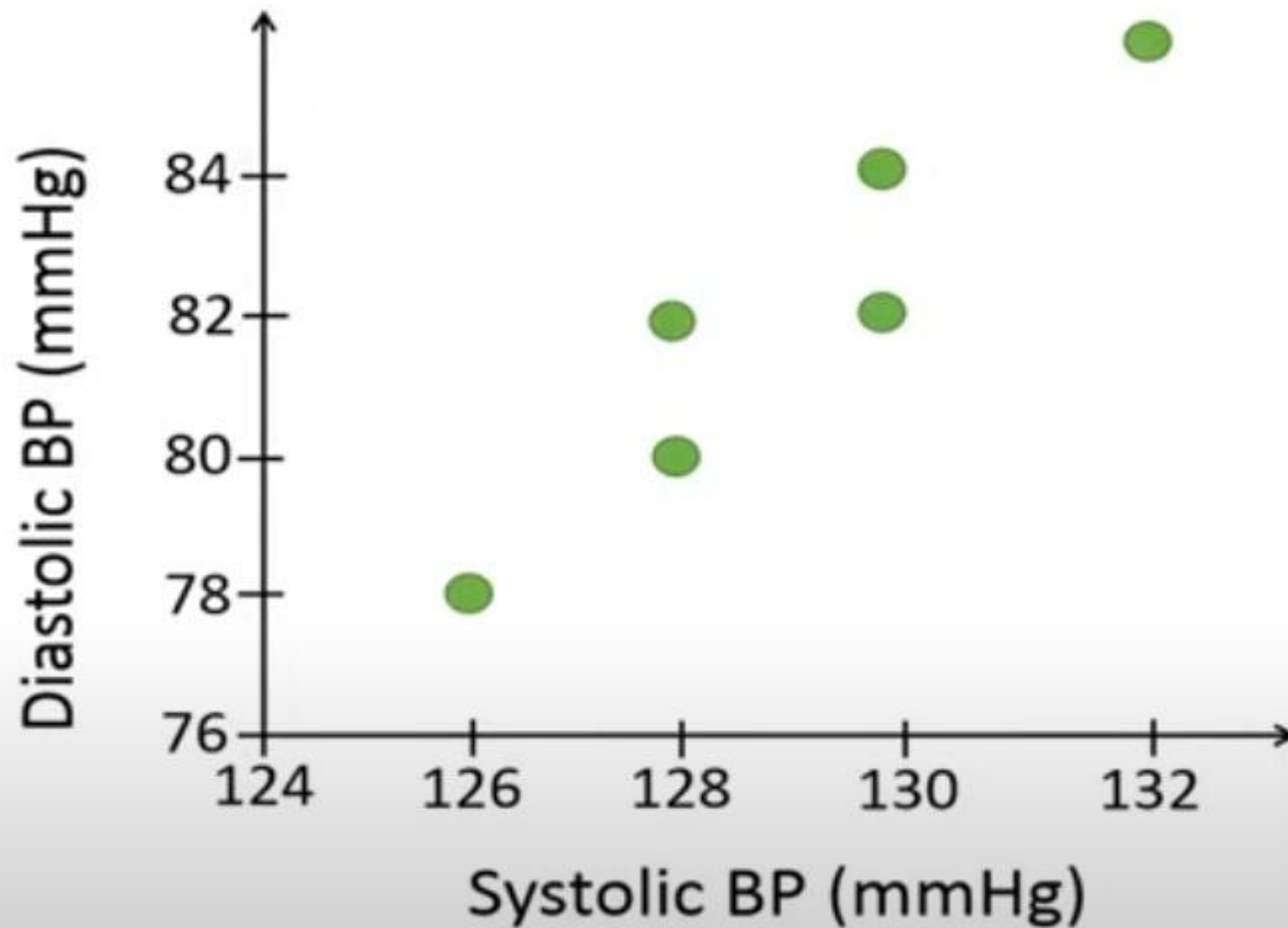
- a_{ii} = the variance of each $\mathbf{F}_i$ (How $\mathbf{F}_i$ correlates with itself). Here $i = 1, 2, 3$.
- $a_{11} + a_{22} + a_{33}$ = Overall variability (total variance).
- $\frac{a_{ii}}{a_{11}+a_{22}+a_{33}} \cdot 100\%$ = Percentage of the total variance that is explained by $\mathbf{F}_i$. Here $i = 1, 2, 3$.

$$\begin{array}{c} \mathbf{F}_1 \\ \mathbf{F}_2 \\ \mathbf{F}_3 \end{array} \begin{bmatrix} \mathbf{F}_1 & \mathbf{F}_2 & \mathbf{F}_3 \\ a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

How to Compute

1. Center the data
2. Calculate the covariance matrix
3. Calculate eigenvalues of the covariance matrix
4. Calculate eigenvectors of the covariance matrix
5. Order the eigenvectors
6. Calculate the principal components

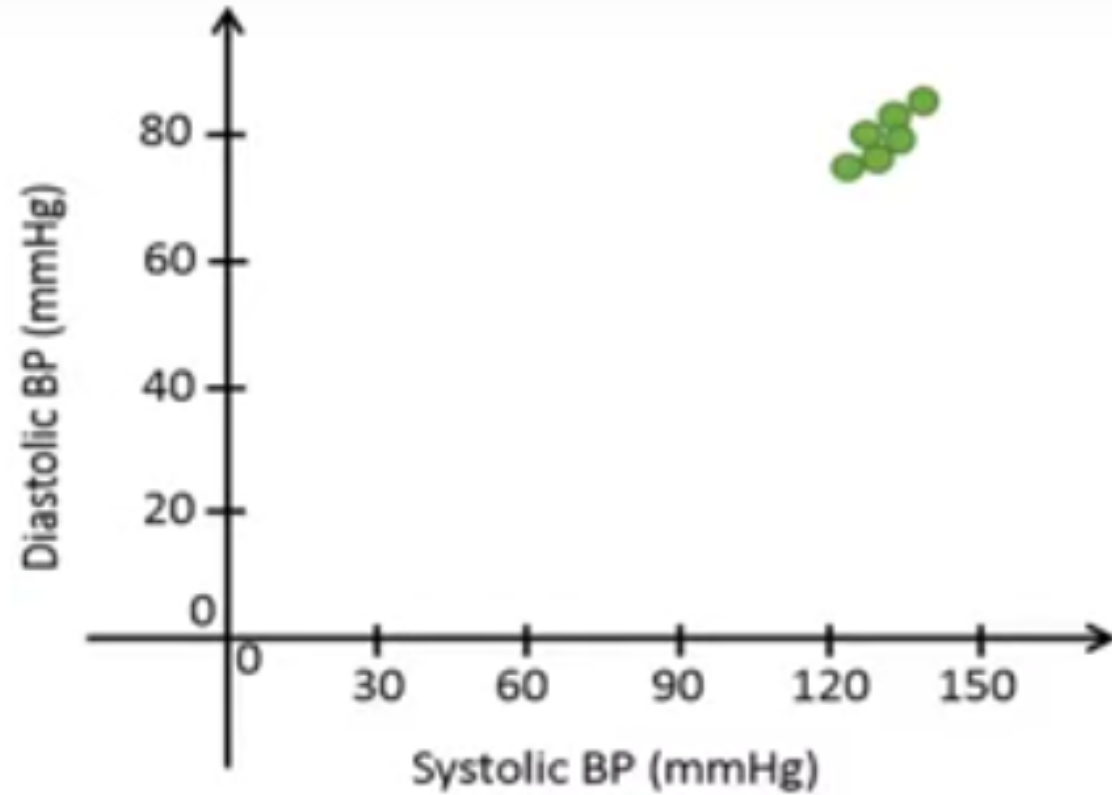
Example



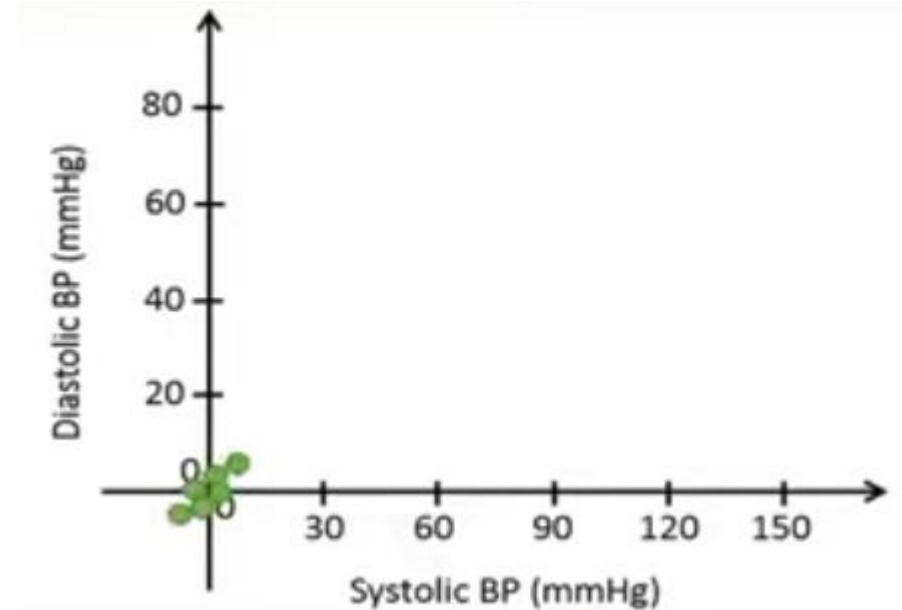
Systolic BP	Diastolic BP
126	78
128	80
128	82
130	82
130	84
132	86



Center Data



Centered SBP	Centered DBP
-3	-4
-1	-2
-1	0
1	0
1	2
3	4



Calculating Covariance Matrix

	SBP	DBP
SBP		
DBP		

$$\text{cov}(x_1, x_2) = \frac{\sum_{i=1}^n (x_{1i} - \bar{x}_1)(x_{2i} - \bar{x}_2)}{n-1}$$

Where:

- $\bar{x}_1$ and $\bar{x}_2$ are the mean values of features x_1 and x_2
- n is the number of data points

The value of covariance can be positive, negative or zeros.

Calculate the eigenvalues of covariance matrix

$$\det|A - \lambda I| = 0$$

$$\det \begin{bmatrix} (4.4 - \lambda) & 5.6 \\ 5.6 & (8.0 - \lambda) \end{bmatrix} = 0$$

$$(4.4 - \lambda)(8.0 - \lambda) - 5.6 \cdot 5.6 = 0$$

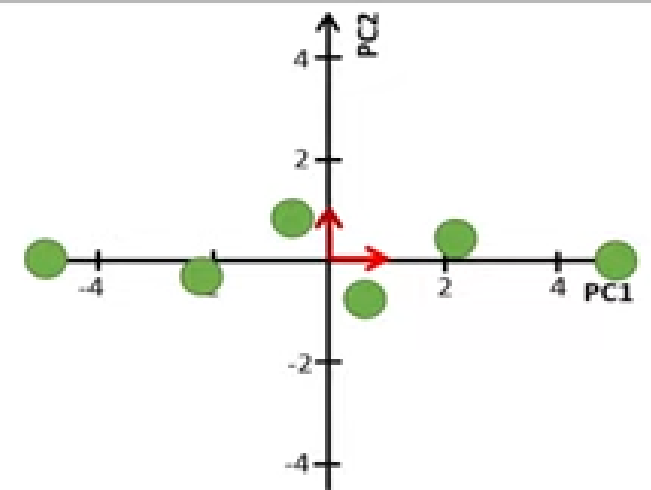
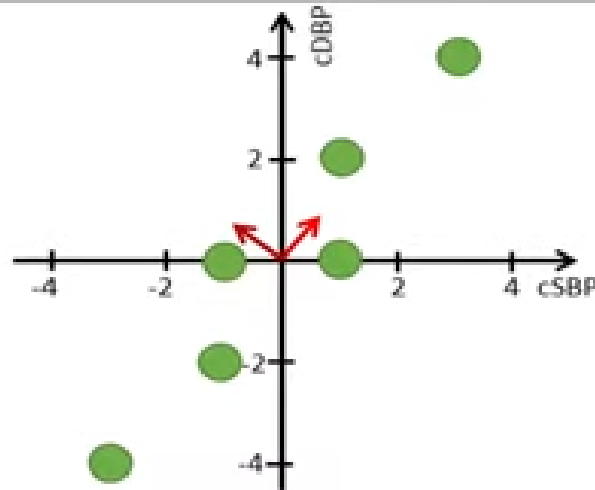
$$3.84 - 12.4\lambda + \lambda^2 = 0$$

$$\lambda_1 = 0.32 \quad \lambda_2 = 12.08$$

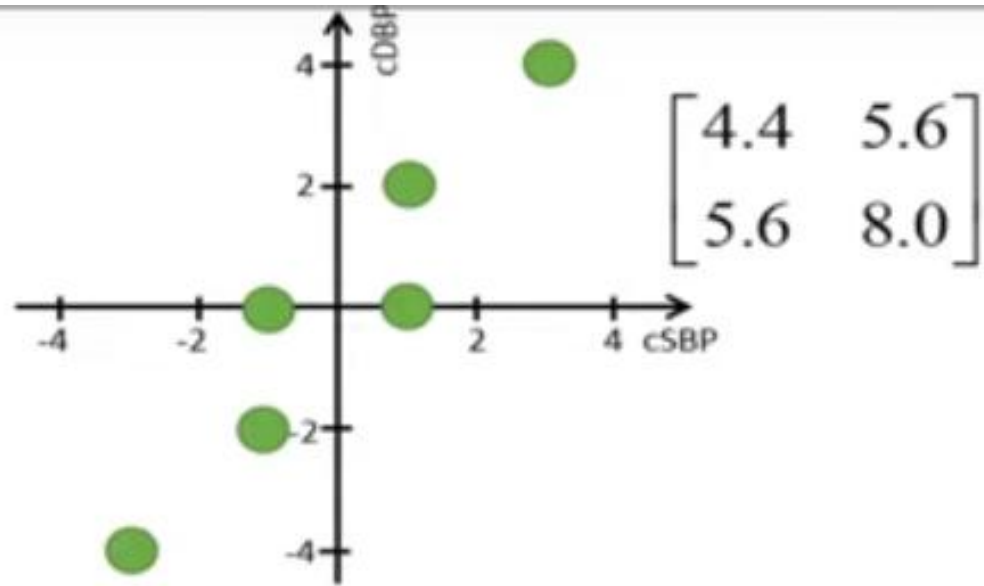
Find Eigenvectors

$$v_2 = \begin{bmatrix} 0.59 \\ 0.81 \end{bmatrix} \quad \lambda_2 = 12.08$$
$$v_1 = \begin{bmatrix} -0.81 \\ 0.59 \end{bmatrix} \quad \lambda_1 = 0.32$$

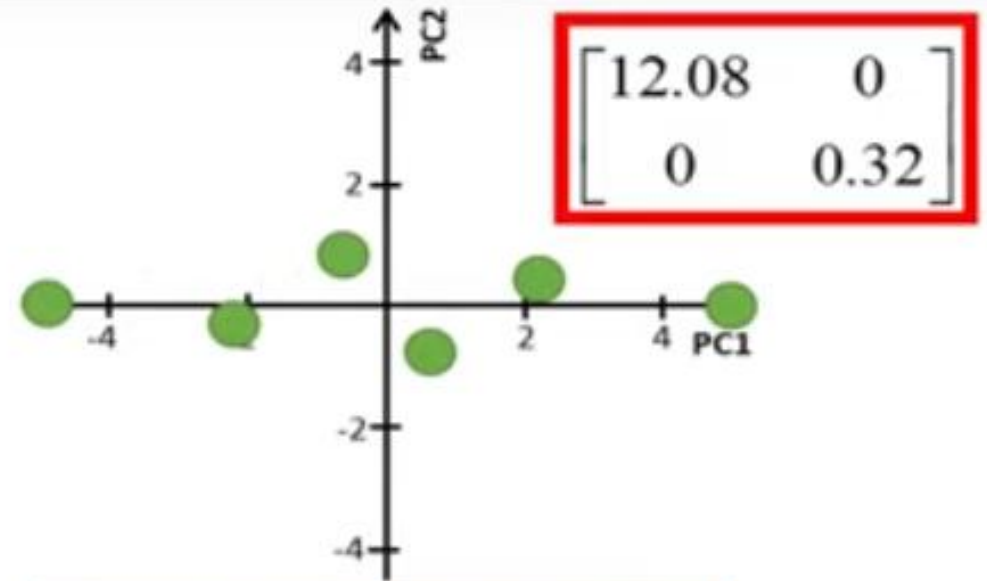
Centered SBP	Centered DBP
-3	-4
-1	-2
-1	0
1	0
1	2
3	4



$$V = \begin{bmatrix} 0.59 & -0.81 \\ 0.81 & 0.59 \end{bmatrix} \quad D = \begin{bmatrix} -3 & -4 \\ -1 & -2 \\ -1 & 0 \\ 1 & 0 \\ 1 & 2 \\ 3 & 4 \end{bmatrix} \quad DV = \begin{bmatrix} -3 & -4 \\ -1 & -2 \\ -1 & 0 \\ 1 & 0 \\ 1 & 2 \\ 3 & 4 \end{bmatrix} \cdot \begin{bmatrix} 0.59 & -0.81 \\ 0.81 & 0.59 \end{bmatrix} = \begin{bmatrix} -5.0 & 0.1 \\ -2.2 & -0.4 \\ -0.6 & 0.8 \\ 0.6 & -0.8 \\ 2.2 & 0.4 \\ 5.0 & -0.1 \end{bmatrix}$$



Centered SBP	Centered DBP
-3	-4
-1	-2
-1	0
1	0
1	2
3	4
Var=4.4	Var=8.0



PC1	PC2
-5.0	0.1
-2.2	-0.4
-0.6	0.8
0.6	-0.8
2.2	0.4
5.0	-0.1
Var=12.08	Var=0.32

$$PC1 = 0.59 \cdot cSBP + 0.81 \cdot cDBP$$

$$PC2 = -0.81 \cdot cSBP + 0.59 \cdot cDBP$$

PC1	PC2
-5.0	0.1
-2.2	-0.4
-0.6	0.8
0.6	-0.7
2.2	0.4
5.0	-0.1
Var=12.08	Var=0.32

